Dictionaries

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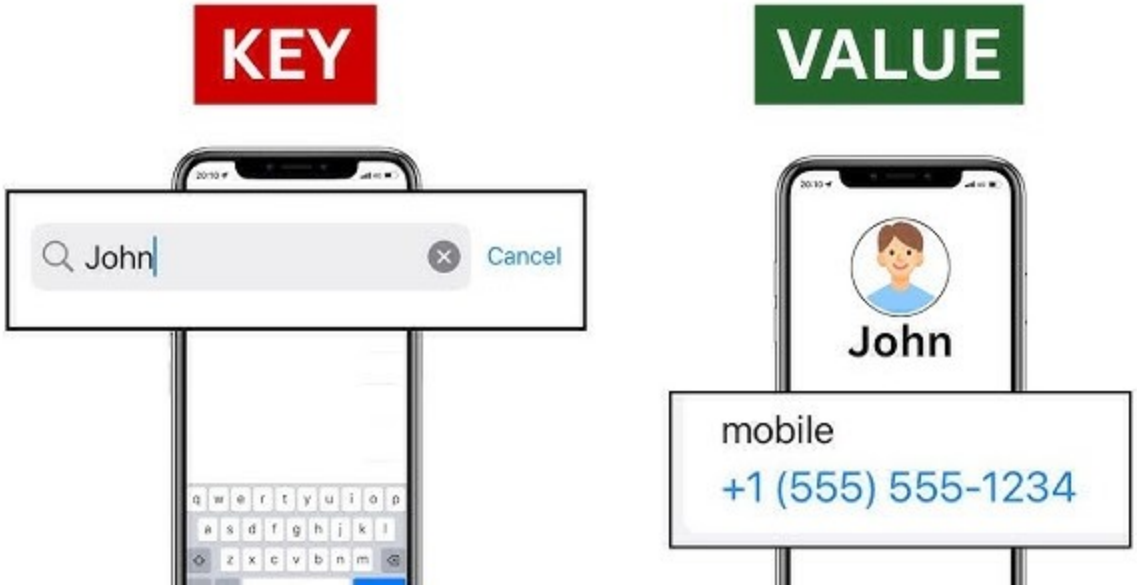
Dictionaries

Dictionaries are used to store data values in key:value pairs.



A dictionary is like a contact list on your phone: each person's name is a **key**, and their phone number is the **value**. You look up the number by using the name, just as you access a value in a dictionary by its key.

Create a Dictionary

Create a dictionary using curly braces `{}`:

```
country_capitals = {
  "Germany": "Berlin",
  "Canada": "Ottawa",
  "England": "London"
}
```

Create an empty dictionary:

```
my_dict = {}
```

or,

```
my_dict = dict()
```

> ⚠️ To create an empty set, you must use the `set()` constructor instead of curly brakets: `my_set = set()`

Access Dictionary Items

Place the **key** inside square brackets `[]` to access the **value** of a dictionary item:

```
country_capitals = {
  "Germany": "Berlin",
  "Canada": "Ottawa",
  "England": "London"
}

# access the value of keys
print(country_capitals["Germany"])    # Output: Berlin
print(country_capitals["England"])    # Output: London
```

Add Items to a Dictionary

We can add an item to a dictionary by assigning a value to a new key.

For example,

```
country_capitals = {
  "Germany": "Berlin",
  "Canada": "Ottawa",
}

# add an item with "Italy" as key and "Rome" as its value
country_capitals["Italy"] = "Rome"

print(country_capitals)
```

Output:

```
{'Germany': 'Berlin', 'Canada': 'Ottawa', 'Italy': 'Rome'}
```

Remove Dictionary Items

We can use the `del` statement to remove an element from a dictionary.

For example,

```
country_capitals = {
  "Germany": "Berlin",
  "Canada": "Ottawa",
}

# delete item having "Germany" key
del country_capitals["Germany"]

print(country_capitals)
```

Output:

```
{'Canada': 'Ottawa'}
```

Change Dictionary Items

Python dictionaries are mutable (changeable). We can change the **value** of a dictionary element by referring to its key.

For example,

```
country_capitals = {
  "Germany": "Berlin",
  "Italy": "Naples",
  "England": "London"
}

# change the value of "Italy" key to "Rome"
country_capitals["Italy"] = "Rome"

print(country_capitals)
```

Output:

```
{'Germany': 'Berlin', 'Italy': 'Rome', 'England': 'London'}
```

Iterate Through a Dictionary

We can iterate through dictionary keys one by one using a for loop.

```
country_capitals = {
    "United States": "Washington D.C.",
    "Italy": "Rome"
}
```

Iterate over the keys of the dictionary:

```
for key in country_capitals:
    print(key)
```

or more explicitly:

```
for key in country_capitals.keys():
    print(key)
```

Output:

```
United States
Italy
```

Iterate over the keys of the dictionary to access their corresponding values.

```
for value in country_capitals.values():
    print(value)
```

Output:

```
Washington D.C.
Rome
```

Iterate over the (key, value) pairs of the dictionary.

```
for key, value in country_capitals.items():
    print(f'{key:<15s}: {value}')
```

Output:

```
United States  : Washington D.C.
Italy          : Rome
```

Reminder: List, Tuple, Sets, and Dictionaries

There are four collection data types in the Python programming language:

- **List** is a collection which is ordered and changeable. Allows duplicate members.
- **Tuple** is a collection which is ordered and unchangeable. Allows duplicate members.
- **Set** is a collection which is unordered, unchangeable*, and unindexed. No duplicate members.
- **Dictionary** is a collection which is ordered** and changeable. No duplicate members.

When choosing a collection type, it is useful to understand the properties of that type. Choosing the right type for a particular data set could mean retention of meaning, and, it could mean an increase in efficiency or security.

Summary Table: Elementary Data Types

Python has the following data types by default:

	Data Type	Class	Example	Mutable	Iterable
Atomic	Integer	`int`	`x = 4`		
	Float	`float`	`x = 3.142`		
	Boolean	`bool`	`x = True`		
Sequence	String	`str`	`x = "this"` or `x = 'this'`		O
	List	`list`	`x = [1, 3.3, 'python']`	O	O
	Tuple	`tuple`	`x = (6, 'me', False)`		O
Collection	Set	`set`	`x = {7, 1, 3, 6, 9}`	O	O
	Dictionary	`dict`	`x = {'a': 1, 'b': 2, 'c': 3}`	O	O

- A **mutable** object is one whose internal state or value can be changed after it has been created.
- An **iterable** is an object that can be "iterated over," meaning it can return its elements one at a time.

Thank You.